KHAHP — Full Technical Evidence & Engineering Register

Public Consolidated Release v5.0 · 19 October 2026 · Sections 01-40 project closeout
Elite Cybersecurity Specialist · Malaysia
200+ hours of owner-reported engineering · 44 security domains · 308 control families · 924 A/B/C assurance checkpoints

> **Engineering mission:** expose enough real implementation detail to demonstrate the quality of our defensive hardening methodology while keeping private host identifiers, secrets and sensitive operational evidence out of the public repository.

1. Executive engineering position and integrity statement

KHAHP is a high-assurance defensive hardening and verification program for a Kicksecure / Debian 13 laptop. The project uses Kicksecure as a **public reference implementation** for the engineering discipline behind Elite Cybersecurity Specialist hardening services.

1.1. Kicksecure OS Royal Hardening — public reference implementation

Kicksecure OS Royal Hardening details are deliberately revealed as an example of the quality, depth and evidence discipline of our work. The public register shows the structure of the investigation, the defensive control families, implementation decisions and sanitized acceptance evidence.

The same evidence-led method is offered for other supported environments, but each target requires its own platform-specific implementation and validation:

Platform / service	Hardening scope offered
Kicksecure OS	Public reference implementation / Royal Hardening engineering demonstration
Ubuntu / Debian Linux	Server and workstation hardening, service reduction, kernel/filesystem/network policy, logging and recovery
RHEL / Rocky / AlmaLinux / CentOS Stream	SELinux-aware hardening, service policy, network controls, provenance and recovery
Windows Server / enterprise workstations	Least privilege, Defender/audit policy, service reduction, network and identity controls

Platform / service	Hardening scope offered
AWS / Azure / Google Cloud / Oracle Cloud	IAM, segmentation, encryption, logging, drift control and recovery assurance
VMware ESXi / Proxmox VE	Management-plane protection, segmentation, update provenance and restore validation
Supported managed endpoints / devices	Vendor-capability-based service, port, update, telemetry and lifecycle hardening

A Kicksecure result does **not** automatically certify the other platforms. Every client environment requires its own scoped controls and evidence.

1.2. Program progress

Workstream	Progress	Current state
Reconnaissance / security review	**40 / 44 sections (90.9%)**	Sections 01-40 completed; Sections 41-44 are the next reconnaissance phase
Hardening implementation	**40 / 44 sections (90.9%)**	Sections 01-40 project-complete; Sections 41-44 have not yet completed reconnaissance or implementation
A/B/C assurance checkpoints	**840 / 924 (90.9%)**	Project-accepted for Sections 01-40
Remaining sections / checkpoints	**4 sections / 84 checkpoints (9.1%)**	Sections 41-44: reconnaissance, implementation and final assurance still to be completed

The **924 checkpoints are not 924 vulnerabilities**. They are 308 control families multiplied by three assurance dimensions: provenance/configuration (A), effective enforcement (B), and persistence/recovery (C). A checkpoint can validate an inherited protection, document a project hardening change, record an accepted compatibility decision, or prove lifecycle behavior.

1.3. What "01-40 complete" means

Sections **01-40 are project-accepted with no remaining implementation gaps** in the current KHAHP master. The 840 associated A/B/C checkpoints are closed as implemented, satisfied, accepted design, accepted compatibility/policy disposition, or not-applicable within project scope. Re-testing is required after material changes such as major updates, kernel changes, policy changes, hardware changes or release-image creation.

This is **project/owner acceptance**, not third-party CIS, DISA or KSPP certification. Section 43 remains the dedicated control-by-control standards-mapping phase, and Section 44 remains final whole-program acceptance for all 44 sections.

1.4. Security finding and acceptance vocabulary

Classification	Meaning
Observed protection	Existing Kicksecure/Debian or project protection inspected or tested
Confirmed security weakness / defect	Demonstrated undesirable behavior against an approved security requirement
Hardening enhancement	Additional defense-in-depth improvement, even without a reproduced exploit
Policy / compatibility disposition	Deliberate design trade-off, exception or N/A decision accepted by the project
Project accepted	Implementation/control family closed within KHAHP scope and current baseline
External certification	Independent certification/assessment by the relevant external authority; **not claimed by this project**

1.5. High-value completed engineering outcomes

Defensive domain	Accepted project outcome
Privilege separation	Distinct USER desktop and privileged SYSMAINT maintenance roles retained; no weakening of USER administrative restrictions for convenience

Defensive domain	Accepted project outcome
Kernel / memory	Restricted kernel pointers, dmesg, ptrace and unprivileged BPF plus hardened boot/memory parameters
Encrypted storage / boot	LUKS root, Secure Boot evidence, signed DKMS evidence and corrected /boot -> /boot/efi ordering
Filesystem	/tmp nodev,nosuid hardening with compatibility-driven noexec disposition
DNS / network / firewall	Local Unbound, DNSSEC, authenticated DoT architecture, nftables and Tor/SOCKS policy reconciled into the hardened baseline
Devices / privacy	USBGuard owner policy, camera/microphone USER denial, Bluetooth block and desktop privacy controls
Browser / application privacy	Firefox privacy controls, seccomp observations and thumbnail-generation hardening
Audit / integrity	auditd active with effective KHAHP rules in normal USER boot; production AIDE scheduled check completed successfully; persistent AIDE lock-helper policy repaired

2. Evidence model and status rules

Gate	Assurance purpose
A - provenance/configuration	Approved source, package/file ownership, intended policy, accepted design or exception
B - effective enforcement	Role-appropriate runtime behavior, bounded allow/deny proof or accepted applicability determination
C - persistence/recovery	Relevant reboot/reconnect/update/recovery behavior or documented project exception

For Sections 01-40 the authoritative private ledger marks all **840 gates as** `PROJECT_ACCEPTED`. Public documentation intentionally summarizes sensitive

evidence rather than publishing raw host identifiers, private rulesets, keys, addresses or confidential logs.

3. Configuration changes, corrective actions and durable results

ID / area	Accepted result
Filesystem / `/tmp`	`nodev,nosuid` retained; blanket `noexec` rejected as an owner-approved compatibility disposition
`/boot` and `/boot/efi`	fstab ordering corrected, immediate validation completed and carried into the accepted baseline
Desktop privacy	Thumbnail generation disabled and persistence verified
Section 40 AIDE	`/usr/bin/dotlockfile` persistent override corrected to `755 root:mail`; production `dailyaidecheck.service` subsequently completed successfully
Section 40 auditd	Persistent KHAHP rules compile cleanly and are effectively loaded in normal USER boot; SYSMAINT no-rules state is role-specific because auditd is inactive there

4. Current project boundary

- **Sections 01-40:** complete / project-accepted / no open implementation gaps.
- **Section 39:** owner-complete and excluded from retest unless the owner intentionally reopens firmware work.
- **Section 40:** completed with final auditd/AIDE acceptance evidence captured on 19 October 2026.
- **Sections 41-44:** remaining program work: persistence/supply chain, secrets/backup/recovery, detailed standards mapping, and final 44-section acceptance.
- **Commercial image/release engineering:** separate from the laptop hardening baseline and remains subject to clean-room/release tests.

5. Implementation review by all 44 sections

Each section retains seven technical control families and 21 A/B/C checkpoint identifiers. For Sections 01-40, the status shown here is **project accepted**. Sections 41-44 remain the active next phase.

01 — Platform / OS / Hardware Baseline

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Debian 13 / Kicksecure platform identity, kernel, hardware, storage and baseline health were reconciled into the authoritative master.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
01.01	Operating-system identity and Debian/Kicksecure release baseline	KHAHP-01.01A · KHAHP-01.01B · KHAHP-01.01C
01.02	Kernel release, architecture and build identity	KHAHP-01.02A · KHAHP-01.02B · KHAHP-01.02C
01.03	CPU model, topology and virtualization capability	KHAHP-01.03A · KHAHP-01.03B · KHAHP-01.03C
01.04	Physical memory and swap capacity baseline	KHAHP-01.04A · KHAHP-01.04B · KHAHP-01.04C
01.05	Storage controller, disk and filesystem topology	KHAHP-01.05A · KHAHP-01.05B · KHAHP-01.05C
01.06	UEFI/bare-metal/virtualization platform state	KHAHP-01.06A · KHAHP-01.06B · KHAHP-01.06C
01.07	Failed-unit, uptime, boot-time and baseline health state	KHAHP-01.07A · KHAHP-01.07B · KHAHP-01.07C

02 — Kicksecure Package Provenance

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Package database health and security-package provenance were reviewed; package-owned and locally modified security configuration is tracked in the private evidence set.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply

third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
02.01	Core Kicksecure package installation and exact versions	KHAHP-02.01A · KHAHP-02.01B · KHAHP-02.01C
02.02	Supporting security-package installation and exact versions	KHAHP-02.02A · KHAHP-02.02B · KHAHP-02.02C
02.03	APT source files and repository configuration	KHAHP-02.03A · KHAHP-02.03B · KHAHP-02.03C
02.04	APT policy and repository origin attribution	KHAHP-02.04A · KHAHP-02.04B · KHAHP-02.04C
02.05	Package ownership of security-critical configuration files	KHAHP-02.05A · KHAHP-02.05B · KHAHP-02.05C
02.06	Package holds, partial installs and dpkg database health	KHAHP-02.06A · KHAHP-02.06B · KHAHP-02.06C
02.07	Local modifications versus package-owned baseline	KHAHP-02.07A · KHAHP-02.07B · KHAHP-02.07C

03 — USER / SYSMAINT Privilege Architecture

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Normal USER and privileged SYSMAINT roles remain separated. Privileged maintenance is performed only in the dedicated maintenance boot context.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
03.01	USER, SYSMAINT and root account separation	KHAHP-03.01A · KHAHP-03.01B · KHAHP-03.01C

Family	Exact technical scope	A/B/C evidence IDs
03.02	Distinct UID/GID and home-directory boundaries	KHAHP-03.02A · KHAHP-03.02B · KHAHP-03.02C
03.03	user-sysmaint-split package and owned policy files	KHAHP-03.03A · KHAHP-03.03B · KHAHP-03.03C
03.04	sudo binary ownership, mode and SYSMAINT-only execution	KHAHP-03.04A · KHAHP-03.04B · KHAHP-03.04C
03.05	sudoers syntax and user-sysmaint- split policy	KHAHP-03.05A · KHAHP-03.05B · KHAHP-03.05C
03.06	USER negative sudo/root acquisition probe	KHAHP-03.06A · KHAHP-03.06B · KHAHP-03.06C
03.07	Login-shell and boot-role separation behavior	KHAHP-03.07A · KHAHP-03.07B · KHAHP-03.07C

04 — SYSMAINT Boot Role

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: SYSMAINT boot markers and role-specific maintenance workflow are retained as part of the hardened operating model.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
04.01	boot-role=sysmaint kernel marker	KHAHP-04.01A · KHAHP-04.01B · KHAHP-04.01C
04.02	systemd.unit=sysmaint-boot.target marker	KHAHP-04.02A · KHAHP-04.02B · KHAHP-04.02C
04.03	sysmaint-boot.target package provenance and unit content	KHAHP-04.03A · KHAHP-04.03B · KHAHP-04.03C

Family	Exact technical scope	A/B/C evidence IDs
04.04	Required SYSMAINT dependencies and wanted units	KHAHP-04.04A · KHAHP-04.04B · KHAHP-04.04C
04.05	Intentionally omitted USER-only or incompatible hardeners	KHAHP-04.05A · KHAHP-04.05B · KHAHP-04.05C
04.06	Conflicting USER/graphical boot-role negative checks	KHAHP-04.06A · KHAHP-04.06B · KHAHP-04.06C
04.07	Failed units and SYSMAINT boot health	KHAHP-04.07A · KHAHP-04.07B · KHAHP-04.07C

05 — Accounts / UID / GID / Local Identity

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Local account, UID/GID, service-account and sensitive identity-file controls were reviewed and accepted for the hardened baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
05.01	root, USER and SYSMAINT account identity	KHAHP-05.01A · KHAHP-05.01B · KHAHP-05.01C
05.02	Duplicate UID and duplicate GID detection	KHAHP-05.02A · KHAHP-05.02B · KHAHP-05.02C
05.03	UID 0 exclusivity and privileged- account review	KHAHP-05.03A · KHAHP-05.03B · KHAHP-05.03C
05.04	Low-UID service-account shell classification	KHAHP-05.04A · KHAHP-05.04B · KHAHP-05.04C
05.05	Home-directory ownership and permissions	KHAHP-05.05A · KHAHP-05.05B · KHAHP-05.05C

Family	Exact technical scope	A/B/C evidence IDs
05.06	shadow/gshadow/passwd/group ownership and modes	KHAHP-05.06A · KHAHP-05.06B · KHAHP-05.06C
05.07	NSS identity source configuration and root lock state	KHAHP-05.07A · KHAHP-05.07B · KHAHP-05.07C

06 — LUKS / Disk / Swap Encryption

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: LUKS-encrypted root storage and encrypted-storage policy are part of the accepted baseline; recovery handling is governed by the project recovery policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
06.01	Root filesystem source and dm-crypt mapping	KHAHP-06.01A · KHAHP-06.01B · KHAHP-06.01C
06.02	LUKS2 backing partition and cryptsetup status	KHAHP-06.02A · KHAHP-06.02B · KHAHP-06.02C
06.03	LUKS header metadata without key- material disclosure	KHAHP-06.03A · KHAHP-06.03B · KHAHP-06.03C
06.04	Root encryption chain from block device to mountpoint	KHAHP-06.04A · KHAHP-06.04B · KHAHP-06.04C
06.05	Active swap inventory and encrypted placement	KHAHP-06.05A · KHAHP-06.05B · KHAHP-06.05C
06.06	swapfile ownership, mode and containing filesystem	KHAHP-06.06A · KHAHP-06.06B · KHAHP-06.06C
06.07	crypttab/fstab encryption persistence and boot mapping	KHAHP-06.07A · KHAHP-06.07B · KHAHP-06.07C

07 — Filesystem / Mount Security

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Filesystem and mount hardening includes /tmp nodev,nosuid and corrected /boot before /boot/efi ordering. noexec exclusions are accepted compatibility decisions, not open defects.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
07.01	Effective mount table and filesystem inventory	KHAHP-07.01A · KHAHP-07.01B · KHAHP-07.01C
07.02	Root mount source, filesystem and options	KHAHP-07.02A · KHAHP-07.02B · KHAHP-07.02C
07.03	/boot and /boot/efi separation, ownership and exposure	KHAHP-07.03A · KHAHP-07.03B · KHAHP-07.03C
07.04	/tmp tmpfs, sticky bit and mount- option review	KHAHP-07.04A · KHAHP-07.04B · KHAHP-07.04C
07.05	/var/tmp sticky-bit and separate- mount review	KHAHP-07.05A · KHAHP-07.05B · KHAHP-07.05C
07.06	/home containment, ownership and inherited options	KHAHP-07.06A · KHAHP-07.06B · KHAHP-07.06C
07.07	Writable mountpoints, fstab and remount-secure review	KHAHP-07.07A · KHAHP-07.07B · KHAHP-07.07C

08 — GRUB / Bootloader Protection

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: GRUB and bootloader protections were reconciled with the project recovery model and accepted for the 01-40 implementation baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
08.01	GRUB package baseline and generated grub.cfg	KHAHP-08.01A · KHAHP-08.01B · KHAHP-08.01C
08.02	grub.cfg ownership, permissions and menu entries	KHAHP-08.02A · KHAHP-08.02B · KHAHP-08.02C
08.03	GRUB password_pbkdf2 and superusers directives	KHAHP-08.03A · KHAHP-08.03B · KHAHP-08.03C
08.04	Kicksecure CPU/kernel/ram-wipe/remount/signed -module fragments	KHAHP-08.04A · KHAHP-08.04B · KHAHP-08.04C
08.05	Quiet-boot and recovery-restriction fragments	KHAHP-08.05A · KHAHP-08.05B · KHAHP-08.05C
08.06	Effective boot command-line correlation	KHAHP-08.06A · KHAHP-08.06B · KHAHP-08.06C
08.07	/boot and EFI exposure versus cryptographic integrity limits	KHAHP-08.07A · KHAHP-08.07B · KHAHP-08.07C

09 — CPU Vulnerability Mitigations

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: CPU vulnerability mitigations and hardened boot parameters are part of the accepted runtime baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
09.01	CPU identity, vendor and mitigation-capability baseline	KHAHP-09.01A · KHAHP-09.01B · KHAHP-09.01C
09.02	SMT/Hyper-Threading disablement and topology	KHAHP-09.02A · KHAHP-09.02B · KHAHP-09.02C

Family	Exact technical scope	A/B/C evidence IDs
09.03	Runtime microcode revision and package baseline	KHAHP-09.03A · KHAHP-09.03B · KHAHP-09.03C
09.04	Kernel vulnerability-interface enumeration	KHAHP-09.04A · KHAHP-09.04B · KHAHP-09.04C
09.05	Spectre/Meltdown/L1TF/MDS/ Retbleed/BHI/SSB/TAA/GDS mitigations	KHAHP-09.05A · KHAHP-09.05B · KHAHP-09.05C
09.06	Mitigation-related kernel command- line parameters	KHAHP-09.06A · KHAHP-09.06B · KHAHP-09.06C
09.07	CPU mitigation flags and kernel-log confirmation	KHAHP-09.07A · KHAHP-09.07B · KHAHP-09.07C

10 — IOMMU / DMA Protection

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: IOMMU / DMA isolation controls and firmware-related boundaries were reconciled with the owner-completed firmware policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
10.01	CPU virtualization/VT-d capability baseline	KHAHP-10.01A · KHAHP-10.01B · KHAHP-10.01C
10.02	IOMMU kernel parameters and passthrough policy	KHAHP-10.02A · KHAHP-10.02B · KHAHP-10.02C
10.03	DMAR/IOMMU runtime enablement	KHAHP-10.03A · KHAHP-10.03B · KHAHP-10.03C
10.04	Translated default domain and strict invalidation	KHAHP-10.04A · KHAHP-10.04B · KHAHP-10.04C

Family	Exact technical scope	A/B/C evidence IDs
10.05	IRQ/interrupt remapping	KHAHP-10.05A · KHAHP-10.05B · KHAHP-10.05C
10.06	IOMMU group topology and full PCI membership	KHAHP-10.06A · KHAHP-10.06B · KHAHP-10.06C
10.07	Early PCI DMA restriction and firmware WSMT evidence	KHAHP-10.07A · KHAHP-10.07B · KHAHP-10.07C

11 — Kernel Sysctl Hardening

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Effective kernel restrictions include kptr_restrict=2, dmesg_restrict=1, ptrace_scope=3 and restricted unprivileged BPF, alongside additional hardened sysctls.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
11.01	Sysctl source-tree inventory and precedence	KHAHP-11.01A · KHAHP-11.01B · KHAHP-11.01C
11.02	Kicksecure security-misc sysctl source files	KHAHP-11.02A · KHAHP-11.02B · KHAHP-11.02C
11.03	Kernel information-leak restrictions	KHAHP-11.03A · KHAHP-11.03B · KHAHP-11.03C
11.04	ASLR, mmap, kexec and privileged-coredump controls	KHAHP-11.04A · KHAHP-11.04B · KHAHP-11.04C
11.05	BPF, ptrace, perf and TTY line- discipline restrictions	KHAHP-11.05A · KHAHP-11.05B · KHAHP-11.05C
11.06	Protected links/FIFOs/regular-file controls	KHAHP-11.06A · KHAHP-11.06B · KHAHP-11.06C

Family	Exact technical scope	A/B/C evidence IDs
11.07	Network-stack sysctl hardening observations	KHAHP-11.07A · KHAHP-11.07B · KHAHP-11.07C

12 — Kernel Boot / Memory Hardening

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Kernel boot and memory-hardening parameters were reconciled as part of the accepted 01-40 baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
12.01	Slab allocator hardening and slab_nomerge	KHAHP-12.01A · KHAHP-12.01B · KHAHP-12.01C
12.02	init_on_alloc and init_on_free memory initialization	KHAHP-12.02A · KHAHP-12.02B · KHAHP-12.02C
12.03	page allocator shuffle and stack randomization	KHAHP-12.03A · KHAHP-12.03B · KHAHP-12.03C
12.04	vsyscall, vdso32 and 32-bit compatibility restrictions	KHAHP-12.04A · KHAHP-12.04B · KHAHP-12.04C
12.05	debugfs, KFENCE and kernel debug- surface controls	KHAHP-12.05A · KHAHP-12.05B · KHAHP-12.05C
12.06	CFI/KCFI and boot-time control acceptance	KHAHP-12.06A · KHAHP-12.06B · KHAHP-12.06C
12.07	pstore/ERST/proc_mem/entropy trust boot parameters	KHAHP-12.07A · KHAHP-12.07B · KHAHP-12.07C

13 — Kernel Module Restrictions

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Kernel-module restriction policy, prohibited/legacy module handling and compatibility exceptions are accepted within project scope.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
13.01	security-misc module blacklist configuration	KHAHP-13.01A · KHAHP-13.01B · KHAHP-13.01C
13.02	Hard-disable install rules for risky modules	KHAHP-13.02A · KHAHP-13.02B · KHAHP-13.02C
13.03	Loaded-module inventory versus policy	KHAHP-13.03A · KHAHP-13.03B · KHAHP-13.03C
13.04	Legacy filesystem module restrictions	KHAHP-13.04A · KHAHP-13.04B · KHAHP-13.04C
13.05	Legacy network/protocol module restrictions	KHAHP-13.05A · KHAHP-13.05B · KHAHP-13.05C
13.06	USB/storage/media module restriction exceptions	KHAHP-13.06A · KHAHP-13.06B · KHAHP-13.06C
13.07	Module-load negative tests and compatibility exceptions	KHAHP-13.07A · KHAHP-13.07B · KHAHP-13.07C

14 — Kernel Module Signing / DKMS

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Secure Boot and signed Tirdad DKMS evidence were reconciled with the module-signing policy and accepted for the current baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
14.01	Loaded-module signer inventory	KHAHP-14.01A · KHAHP-14.01B · KHAHP-14.01C
14.02	DKMS package and built-module inventory	KHAHP-14.02A · KHAHP-14.02B · KHAHP-14.02C
14.03	tirdad DKMS build, signer and runtime state	KHAHP-14.03A · KHAHP-14.03B · KHAHP-14.03C
14.04	NVIDIA/other third-party module signer state	KHAHP-14.04A · KHAHP-14.04B · KHAHP-14.04C
14.05	Kernel module signature enforcement capability	KHAHP-14.05A · KHAHP-14.05B · KHAHP-14.05C
14.06	MOK/enrolled certificate readiness	KHAHP-14.06A · KHAHP-14.06B · KHAHP-14.06C
14.07	Secure-Boot compatibility and unsigned-module exceptions	KHAHP-14.07A · KHAHP-14.07B · KHAHP-14.07C

15 — AppArmor / Mandatory Access Control

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: AppArmor is enabled broadly. Firefox retains the vendor unconfined placeholder as an explicit compatibility disposition; browser seccomp/privacy controls are retained. This is a documented project exception, not an open implementation gap.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
15.01	AppArmor kernel/module enablement	KHAHP-15.01A · KHAHP-15.01B · KHAHP-15.01C
15.02	Profile inventory and parser status	KHAHP-15.02A · KHAHP-15.02B · KHAHP-15.02C

Family	Exact technical scope	A/B/C evidence IDs
15.03	Enforce/complain/unconfined process counts	KHAHP-15.03A · KHAHP-15.03B · KHAHP-15.03C
15.04	Tor and network-service profile enforcement	KHAHP-15.04A · KHAHP-15.04B · KHAHP-15.04C
15.05	Browser/application profile coverage	KHAHP-15.05A · KHAHP-15.05B · KHAHP-15.05C
15.06	Denied-event review and false-positive handling	KHAHP-15.06A · KHAHP-15.06B · KHAHP-15.06C
15.07	USER versus SYSMAINT AppArmor behavior	KHAHP-15.07A · KHAHP-15.07B · KHAHP-15.07C

16 — SUID / SGID / File Capabilities

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: SUID/SGID binaries and file capabilities were inventoried and reconciled against package ownership and compatibility requirements.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
16.01	Complete SUID file inventory	KHAHP-16.01A · KHAHP-16.01B · KHAHP-16.01C
16.02	Complete SGID file inventory	KHAHP-16.02A · KHAHP-16.02B · KHAHP-16.02C
16.03	Package ownership of privileged binaries	KHAHP-16.03A · KHAHP-16.03B · KHAHP-16.03C
16.04	permission-hardener expected mode validation	KHAHP-16.04A · KHAHP-16.04B · KHAHP-16.04C
16.05	File-capability inventory and package attribution	KHAHP-16.05A · KHAHP-16.05B · KHAHP-16.05C

Family	Exact technical scope	A/B/C evidence IDs
16.06	Unexpected privileged executable detection	KHAHP-16.06A · KHAHP-16.06B · KHAHP-16.06C
16.07	Privilege-binary negative/compatibility review	KHAHP-16.07A · KHAHP-16.07B · KHAHP-16.07C

17 — Permissions / ACL / Umask

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Permissions, ACL, umask and shared-directory controls were reconciled, including vendor-specific exceptions accepted by policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
17.01	System file ownership baseline	KHAHP-17.01A · KHAHP-17.01B · KHAHP-17.01C
17.02	World-writable system path inventory	KHAHP-17.02A · KHAHP-17.02B · KHAHP-17.02C
17.03	Unexpected group-writable sensitive paths	KHAHP-17.03A · KHAHP-17.03B · KHAHP-17.03C
17.04	ACL inventory on sensitive configuration	KHAHP-17.04A · KHAHP-17.04B · KHAHP-17.04C
17.05	Root and user umask policy	KHAHP-17.05A · KHAHP-17.05B · KHAHP-17.05C
17.06	Home and evidence-directory permissions	KHAHP-17.06A · KHAHP-17.06B · KHAHP-17.06C
17.07	Sticky-bit and shared-directory permission policy	KHAHP-17.07A · KHAHP-17.07B · KHAHP-17.07C

18 — PAM / Login / Faillock

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: PAM, yescrypt and login/lockout policy were reconciled with safe recovery requirements and accepted for the implementation baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
18.01	PAM stack package provenance	KHAHP-18.01A · KHAHP-18.01B · KHAHP-18.01C
18.02	password quality and hashing policy	KHAHP-18.02A · KHAHP-18.02B · KHAHP-18.02C
18.03	faillock/brute-force lockout policy	KHAHP-18.03A · KHAHP-18.03B · KHAHP-18.03C
18.04	login delay and retry controls	KHAHP-18.04A · KHAHP-18.04B · KHAHP-18.04C
18.05	su/sudo/polkit PAM interaction	KHAHP-18.05A · KHAHP-18.05B · KHAHP-18.05C
18.06	account aging and inactive-account policy	KHAHP-18.06A · KHAHP-18.06B · KHAHP-18.06C
18.07	USER/SYSMAINT login regression tests	KHAHP-18.07A · KHAHP-18.07B · KHAHP-18.07C

19 — SSH Client / Server Security

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Network SSH service exposure remains disabled in the normal USER baseline; client-side/key policy and local uses were separately classified.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
19.01	Local SSH server service/socket state	KHAHP-19.01A · KHAHP-19.01B · KHAHP-19.01C
19.02	sshd listener and inbound port absence	KHAHP-19.02A · KHAHP-19.02B · KHAHP-19.02C
19.03	sshd_config security-misc policy if installed	KHAHP-19.03A · KHAHP-19.03B · KHAHP-19.03C
19.04	SSH client configuration and host-key verification	KHAHP-19.04A · KHAHP-19.04B · KHAHP-19.04C
19.05	Private-key ownership and permission baseline	KHAHP-19.05A · KHAHP-19.05B · KHAHP-19.05C
19.06	Password-authentication and root-login policy for future server use	KHAHP-19.06A · KHAHP-19.06B · KHAHP-19.06C
19.07	Known-hosts/key algorithm and crypto-policy review	KHAHP-19.07A · KHAHP-19.07B · KHAHP-19.07C

20 — Enabled / Running Service Audit

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Enabled/running services, sockets and listeners were reviewed per boot role; the accepted baseline avoids indiscriminate removal of required vendor services.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
20.01	Enabled systemd service inventory	KHAHP-20.01A · KHAHP-20.01B · KHAHP-20.01C
20.02	Running systemd service inventory	KHAHP-20.02A · KHAHP-20.02B · KHAHP-20.02C

Family	Exact technical scope	A/B/C evidence IDs
20.03	Socket/path/timer activated service inventory	KHAHP-20.03A · KHAHP-20.03B · KHAHP-20.03C
20.04	Kicksecure-required service attribution	KHAHP-20.04A · KHAHP-20.04B · KHAHP-20.04C
20.05	Unexpected or unnecessary service classification	KHAHP-20.05A · KHAHP-20.05B · KHAHP-20.05C
20.06	Listener-to-service ownership correlation	KHAHP-20.06A · KHAHP-20.06B · KHAHP-20.06C
20.07	Reboot persistence and failed-service review	KHAHP-20.07A · KHAHP-20.07B · KHAHP-20.07C

21 — systemd Service Sandboxing

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: systemd service-sandboxing controls and compatibility limits were reconciled and accepted for the current service set.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
21.01	systemd-analyze security baseline	KHAHP-21.01A · KHAHP-21.01B · KHAHP-21.01C
21.02	NoNewPrivileges adoption	KHAHP-21.02A · KHAHP-21.02B · KHAHP-21.02C
21.03	ProtectSystem/ProtectHome restrictions	KHAHP-21.03A · KHAHP-21.03B · KHAHP-21.03C
21.04	PrivateTmp/PrivateDevices isolation	KHAHP-21.04A · KHAHP-21.04B · KHAHP-21.04C
21.05	CapabilityBoundingSet and ambient capabilities	KHAHP-21.05A · KHAHP-21.05B · KHAHP-21.05C

Family	Exact technical scope	A/B/C evidence IDs
21.06	RestrictNamespaces/SystemCallFilter/address-family controls	KHAHP-21.06A · KHAHP-21.06B · KHAHP-21.06C
21.07	Service-specific compatibility exceptions and regressions	KHAHP-21.07A · KHAHP-21.07B · KHAHP-21.07C

22 — System.map / Core Dumps / PStore

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Core-dump, pstore and crash-data privacy controls were reconciled with the owner-approved forensic/privacy trade-off.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
22.01	System.map exposure and removal policy	KHAHP-22.01A · KHAHP-22.01B · KHAHP-22.01C
22.02	systemd-coredump configuration	KHAHP-22.02A · KHAHP-22.02B · KHAHP-22.02C
22.03	kernel.core_pattern effective state	KHAHP-22.03A · KHAHP-22.03B · KHAHP-22.03C
22.04	fs.suid_dumpable effective state	KHAHP-22.04A · KHAHP-22.04B · KHAHP-22.04C
22.05	pstore retention/disable policy	KHAHP-22.05A · KHAHP-22.05B · KHAHP-22.05C
22.06	EFI pstore and ERST boot controls	KHAHP-22.06A · KHAHP-22.06B · KHAHP-22.06C
22.07	Crash-evidence privacy versus diagnostic requirements	KHAHP-22.07A · KHAHP-22.07B · KHAHP-22.07C

23 — Journald / Logging / Logrotate

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Journal ownership, access, retention and time-ordering requirements were reconciled into the accepted logging baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
23.01	journald security-misc configuration	KHAHP-23.01A · KHAHP-23.01B · KHAHP-23.01C
23.02	Persistent versus volatile journal storage	KHAHP-23.02A · KHAHP-23.02B · KHAHP-23.02C
23.03	Journal file permissions and ownership	KHAHP-23.03A · KHAHP-23.03B · KHAHP-23.03C
23.04	Rate limiting and size/retention policy	KHAHP-23.04A · KHAHP-23.04B · KHAHP-23.04C
23.05	Forwarding/syslog integration state	KHAHP-23.05A · KHAHP-23.05B · KHAHP-23.05C
23.06	logrotate configuration and sensitive-log modes	KHAHP-23.06A · KHAHP-23.06B · KHAHP-23.06C
23.07	Log integrity, time coherence and reboot persistence	KHAHP-23.07A · KHAHP-23.07B · KHAHP-23.07C

24 — Panic / Recovery / Emergency Shutdown

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Panic, emergency shutdown and controlled recovery behavior were reconciled with owner-approved recovery requirements.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
24.01	kernel panic-on-oops policy	KHAHP-24.01A · KHAHP-24.01B · KHAHP-24.01C
24.02	panic timeout and fail-closed behavior	KHAHP-24.02A · KHAHP-24.02B · KHAHP-24.02C
24.03	emerg-shutdown.service provenance and state	KHAHP-24.03A · KHAHP-24.03B · KHAHP-24.03C
24.04	rescue/emergency target restrictions	KHAHP-24.04A · KHAHP-24.04B · KHAHP-24.04C
24.05	rd.emergency=halt and rd.shell=0 behavior	KHAHP-24.05A · KHAHP-24.05B · KHAHP-24.05C
24.06	Magic SysRq and recovery-surface policy	KHAHP-24.06A · KHAHP-24.06B · KHAHP-24.06C
24.07	Controlled failure/recovery evidence and rollback	KHAHP-24.07A · KHAHP-24.07B · KHAHP-24.07C

25 — RAM Wipe

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Swap, RAM-wipe and shutdown-memory protections were reconciled with platform capability and recovery policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
25.01	ram-wipe package provenance and version	KHAHP-25.01A · KHAHP-25.01B · KHAHP-25.01C
25.02	GRUB/package RAM-wipe source presence	KHAHP-25.02A · KHAHP-25.02B · KHAHP-25.02C
25.03	init_on_alloc/init_on_free live correlation	KHAHP-25.03A · KHAHP-25.03B · KHAHP-25.03C

Family	Exact technical scope	A/B/C evidence IDs
25.04	Shutdown/reboot wipe hook inventory	KHAHP-25.04A · KHAHP-25.04B · KHAHP-25.04C
25.05	Swap/LUKS interaction during shutdown	KHAHP-25.05A · KHAHP-25.05B · KHAHP-25.05C
25.06	Controlled reboot/shutdown persistence test	KHAHP-25.06A · KHAHP-25.06B · KHAHP-25.06C
25.07	Limitations of software RAM wiping and cold-boot scope	KHAHP-25.07A · KHAHP-25.07B · KHAHP-25.07C

26 — USBGuard / Peripheral Security

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: USBGuard policy retains authorized keyboards, mice and recovery/removable storage while controlling unapproved device classes according to owner policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
26.01	usbguard daemon and DBus service state	KHAHP-26.01A · KHAHP-26.01B · KHAHP-26.01C
26.02	Policy files and rule provenance	KHAHP-26.02A · KHAHP-26.02B · KHAHP-26.02C
26.03	ImplicitPolicyTarget and device-policy settings	KHAHP-26.03A · KHAHP-26.03B · KHAHP-26.03C
26.04	Allowed keyboard/mouse/audio/video/storage classes	KHAHP-26.04A · KHAHP-26.04B · KHAHP-26.04C
26.05	Rejected HID+extra and RNDIS/network-device patterns	KHAHP-26.05A · KHAHP-26.05B · KHAHP-26.05C

Family	Exact technical scope	A/B/C evidence IDs
26.06	IPC authorization/notifier behavior	KHAHP-26.06A · KHAHP-26.06B · KHAHP-26.06C
26.07	Hot-plug negative tests and reboot persistence	KHAHP-26.07A · KHAHP-26.07B · KHAHP-26.07C

27 — NetworkManager / Wi-Fi / MAC / DHCP

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Network identity and MAC/DHCP privacy controls were reconciled for the accepted operating model.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
27.01	NetworkManager package/service state	KHAHP-27.01A · KHAHP-27.01B · KHAHP-27.01C
27.02	Saved Wi-Fi profile inventory and autoconnect	KHAHP-27.02A · KHAHP-27.02B · KHAHP-27.02C
27.03	Active-profile cloned-MAC and DHCP privacy settings	KHAHP-27.03A · KHAHP-27.03B · KHAHP-27.03C
27.04	Scan MAC randomization	KHAHP-27.04A · KHAHP-27.04B · KHAHP-27.04C
27.05	Ethernet MAC randomization policy	KHAHP-27.05A · KHAHP-27.05B · KHAHP-27.05C
27.06	Runtime MAC versus permanent hardware MAC	KHAHP-27.06A · KHAHP-27.06B · KHAHP-27.06C
27.07	DHCP client identifier and profile persistence	KHAHP-27.07A · KHAHP-27.07B · KHAHP-27.07C

28 — IPv6 Policy

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: IPv6 policy is explicit in the hardened baseline; profile behavior and DNS interactions are treated according to the selected operating mode.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
28.01	ipv6.disable=1 boot parameter	KHAHP-28.01A · KHAHP-28.01B · KHAHP-28.01C
28.02	Absence of IPv6 addresses	KHAHP-28.02A · KHAHP-28.02B · KHAHP-28.02C
28.03	Absence of IPv6 routes	KHAHP-28.03A · KHAHP-28.03B · KHAHP-28.03C
28.04	NetworkManager IPv6 method/state	KHAHP-28.04A · KHAHP-28.04B · KHAHP-28.04C
28.05	Resolver absence of IPv6 nameservers	KHAHP-28.05A · KHAHP-28.05B · KHAHP-28.05C
28.06	IPv6 TCP/UDP listener exposure	KHAHP-28.06A · KHAHP-28.06B · KHAHP-28.06C
28.07	Reboot persistence and DNS/HTTPS regression test	KHAHP-28.07A · KHAHP-28.07B · KHAHP-28.07C

29 — Unbound / Local DNS Architecture

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Unbound is the accepted local resolver path in normal USER mode, with role-specific service semantics reconciled.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
29.01	Unbound package/service state	KHAHP-29.01A · KHAHP-29.01B · KHAHP-29.01C
29.02	127.0.0.1:53 UDP/TCP listener and non-loopback absence	KHAHP-29.02A · KHAHP-29.02B · KHAHP-29.02C
29.03	resolv.conf local-resolver wiring	KHAHP-29.03A · KHAHP-29.03B · KHAHP-29.03C
29.04	NetworkManager effective DNS assignment	KHAHP-29.04A · KHAHP-29.04B · KHAHP-29.04C
29.05	Unbound configuration include tree validation	KHAHP-29.05A · KHAHP-29.05B · KHAHP-29.05C
29.06	DoT forward-zone and IPv6 source policy	KHAHP-29.06A · KHAHP-29.06B · KHAHP-29.06C
29.07	USER/SYSMAINT service behavior and persistence	KHAHP-29.07A · KHAHP-29.07B · KHAHP-29.07C

30 — DNSSEC / DoT / DNS Leak Prevention

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: DNSSEC validation, authenticated DNS-over-TLS configuration, direct DNS egress policy and browser DNS behavior were reconciled into the accepted DNS architecture.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
30.01	DNSSEC positive validation test	KHAHP-30.01A · KHAHP-30.01B · KHAHP-30.01C
30.02	Broken-DNSSEC negative test	KHAHP-30.02A · KHAHP-30.02B · KHAHP-30.02C

Family	Exact technical scope	A/B/C evidence IDs
30.03	DoT forwarding configuration and reachability scope	KHAHP-30.03A · KHAHP-30.03B · KHAHP-30.03C
30.04	Plaintext external UDP/53 block	KHAHP-30.04A · KHAHP-30.04B · KHAHP-30.04C
30.05	Plaintext external TCP/53 block	KHAHP-30.05A · KHAHP-30.05B · KHAHP-30.05C
30.06	Local DNS functionality after firewall enforcement	KHAHP-30.06A · KHAHP-30.06B · KHAHP-30.06C
30.07	DNS leak/bypass and reboot-persistence testing	KHAHP-30.07A · KHAHP-30.07B · KHAHP-30.07C

31 — nftables Firewall

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: nftables policy and persistence are part of the accepted firewall baseline, including the project DNS-egress restrictions.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
31.01	nftables package/service/runtime ruleset state	KHAHP-31.01A · KHAHP-31.01B · KHAHP-31.01C
31.02	Dedicated inet kicksecure_hardening table	KHAHP-31.02A · KHAHP-31.02B · KHAHP-31.02C
31.03	Input default-drop policy	KHAHP-31.03A · KHAHP-31.03B · KHAHP-31.03C
31.04	Forward default-drop policy	KHAHP-31.04A · KHAHP-31.04B · KHAHP-31.04C
31.05	Loopback, established/related and invalid handling	KHAHP-31.05A · KHAHP-31.05B · KHAHP-31.05C

Family	Exact technical scope	A/B/C evidence IDs
31.06	DHCP/ICMP and required inbound exceptions	KHAHP-31.06A · KHAHP-31.06B · KHAHP-31.06C
31.07	Static syntax, reload safety and reboot persistence	KHAHP-31.07A · KHAHP-31.07B · KHAHP-31.07C

32 — LAN Discovery / Multicast / LLMNR / mDNS

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Local discovery exposure was reviewed and reconciled against the selected privacy posture.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
32.01	Avahi/mDNS daemon state	KHAHP-32.01A · KHAHP-32.01B · KHAHP-32.01C
32.02	LLMNR/resolved multicast name-resolution state	KHAHP-32.02A · KHAHP-32.02B · KHAHP-32.02C
32.03	wsdd/Samba discovery state	KHAHP-32.03A · KHAHP-32.03B · KHAHP-32.03C
32.04	rpcbind/minidlna/UPnP service state	KHAHP-32.04A · KHAHP-32.04B · KHAHP-32.04C
32.05	Multicast/discovery TCP+UDP listener inventory	KHAHP-32.05A · KHAHP-32.05B · KHAHP-32.05C
32.06	Firewall exposure of discovery ports	KHAHP-32.06A · KHAHP-32.06B · KHAHP-32.06C
32.07	Negative LAN discovery and reboot-persistence test	KHAHP-32.07A · KHAHP-32.07B · KHAHP-32.07C

33 — Tor / SOCKS / DNS Isolation

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Tor / SOCKS routing policy and intended application scope were reconciled with the project network-privacy model.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
33.01	Tor package/service state	KHAHP-33.01A · KHAHP-33.01B · KHAHP-33.01C
33.02	SOCKS listener 127.0.0.1:9050 and non-loopback absence	KHAHP-33.02A · KHAHP-33.02B · KHAHP-33.02C
33.03	Tor configuration provenance and syntax validation	KHAHP-33.03A · KHAHP-33.03B · KHAHP-33.03C
33.04	Application proxy routing where intended	KHAHP-33.04A · KHAHP-33.04B · KHAHP-33.04C
33.05	Tor DNS isolation behavior	KHAHP-33.05A · KHAHP-33.05B · KHAHP-33.05C
33.06	Clearnet bypass classification for Tor-mode applications	KHAHP-33.06A · KHAHP-33.06B · KHAHP-33.06C
33.07	Bootstrap, circuit, restart and failure-mode behavior	KHAHP-33.07A · KHAHP-33.07B · KHAHP-33.07C

34 — sdwdate / Secure Time / Boot Clock

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Secure-time services and chronology behavior were reconciled into the accepted baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
34.01	sdwdate package/service state	KHAHP-34.01A · KHAHP-34.01B · KHAHP-34.01C
34.02	timesanitycheck execution state	KHAHP-34.02A · KHAHP-34.02B · KHAHP-34.02C
34.03	bootclockrandomization execution state	KHAHP-34.03A · KHAHP-34.03B · KHAHP-34.03C
34.04	System clock sanity metadata	KHAHP-34.04A · KHAHP-34.04B · KHAHP-34.04C
34.05	NTP/systemd-timesyncd conflicts	KHAHP-34.05A · KHAHP-34.05B · KHAHP-34.05C
34.06	USER versus SYSMAINT time-source behavior	KHAHP-34.06A · KHAHP-34.06B · KHAHP-34.06C
34.07	Time rollback/jump detection and log coherence	KHAHP-34.07A · KHAHP-34.07B · KHAHP-34.07C

35 — Entropy / RNG

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Entropy, random-seed and early-boot randomness controls were reconciled as part of the accepted cryptographic baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
35.01	Kernel RNG readiness at boot	KHAHP-35.01A · KHAHP-35.01B · KHAHP-35.01C
35.02	haveged package/service state	KHAHP-35.02A · KHAHP-35.02B · KHAHP-35.02C
35.03	jitterentropy-rngd package/service state	KHAHP-35.03A · KHAHP-35.03B · KHAHP-35.03C

Family	Exact technical scope	A/B/C evidence IDs
35.04	random.trust_cpu boot policy	KHAHP-35.04A · KHAHP-35.04B · KHAHP-35.04C
35.05	random.trust_bootloader boot policy	KHAHP-35.05A · KHAHP-35.05B · KHAHP-35.05C
35.06	Entropy source and /dev/random behavior	KHAHP-35.06A · KHAHP-35.06B · KHAHP-35.06C
35.07	Early-boot entropy persistence and regression evidence	KHAHP-35.07A · KHAHP-35.07B · KHAHP-35.07C

36 — Browser DoH / WebRTC / Telemetry

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Firefox DoH, WebRTC, telemetry, prefetch and related privacy controls were reconciled and accepted for the hardened browser baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
36.01	Browser DNS-over-HTTPS/TRR setting	KHAHP-36.01A · KHAHP-36.01B · KHAHP-36.01C
36.02	Browser proxy DNS routing behavior	KHAHP-36.02A · KHAHP-36.02B · KHAHP-36.02C
36.03	WebRTC/STUN local/public IP exposure	KHAHP-36.03A · KHAHP-36.03B · KHAHP-36.03C
36.04	Telemetry and crash-report settings	KHAHP-36.04A · KHAHP-36.04B · KHAHP-36.04C
36.05	Safe browsing/prediction/background network features	KHAHP-36.05A · KHAHP-36.05B · KHAHP-36.05C
36.06	Extension/add-on inventory and trust	KHAHP-36.06A · KHAHP-36.06B · KHAHP-36.06C

Family	Exact technical scope	A/B/C evidence IDs
36.07	Browser update, profile persistence and leak tests	KHAHP-36.07A · KHAHP-36.07B · KHAHP-36.07C

37 — Application / File / Metadata Privacy

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Thumbnail generation is disabled and persisted across reboot; application-history/metadata handling was reconciled into the accepted desktop privacy baseline.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
37.01	Desktop search/indexing services	KHAHP-37.01A · KHAHP-37.01B · KHAHP-37.01C
37.02	Thumbnail and preview cache behavior	KHAHP-37.02A · KHAHP-37.02B · KHAHP-37.02C
37.03	Recent-files/history and clipboard retention	KHAHP-37.03A · KHAHP-37.03B · KHAHP-37.03C
37.04	Document/image metadata handling	KHAHP-37.04A · KHAHP-37.04B · KHAHP-37.04C
37.05	Temporary-file and application-cache exposure	KHAHP-37.05A · KHAHP-37.05B · KHAHP-37.05C
37.06	Sandboxed versus unsandboxed application inventory	KHAHP-37.06A · KHAHP-37.06B · KHAHP-37.06C
37.07	Metadata-removal workflow and regression testing	KHAHP-37.07A · KHAHP-37.07B · KHAHP-37.07C

38 — Camera / Microphone / Power / Screen Lock

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: Camera/microphone USER denial, Bluetooth blocking, screen-lock behavior and peripheral privacy controls were reconciled under owner policy.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
38.01	Camera hardware/device exposure	KHAHP-38.01A · KHAHP-38.01B · KHAHP-38.01C
38.02	Microphone hardware/device exposure	KHAHP-38.02A · KHAHP-38.02B · KHAHP-38.02C
38.03	Bluetooth/rfkill interaction with peripherals	KHAHP-38.03A · KHAHP-38.03B · KHAHP-38.03C
38.04	Screen-lock timeout and authentication	KHAHP-38.04A · KHAHP-38.04B · KHAHP-38.04C
38.05	Suspend/lock-on-lid and idle behavior	KHAHP-38.05A · KHAHP-38.05B · KHAHP-38.05C
38.06	Hibernation configuration and encrypted-swap implications	KHAHP-38.06A · KHAHP-38.06B · KHAHP-38.06C
38.07	Shutdown-versus-suspend operational policy	KHAHP-38.07A · KHAHP-38.07B · KHAHP-38.07C

39 — Firmware / BIOS / Secure Boot / TPM

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: OWNER-COMPLETE / NO RETEST. BIOS/UEFI work is retained as a completed owner-controlled domain; Secure Boot evidence remains part of the project record.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
39.01	BIOS/UEFI vendor, model and firmware version	KHAHP-39.01A · KHAHP-39.01B · KHAHP-39.01C
39.02	fwupd device/security-attribute inventory	KHAHP-39.02A · KHAHP-39.02B · KHAHP-39.02C
39.03	Secure Boot state and Setup Mode	KHAHP-39.03A · KHAHP-39.03B · KHAHP-39.03C
39.04	TPM presence, version and ownership state	KHAHP-39.04A · KHAHP-39.04B · KHAHP-39.04C
39.05	BIOS/UEFI administrator password manual check	KHAHP-39.05A · KHAHP-39.05B · KHAHP-39.05C
39.06	PXE/CSM/external-boot/manual boot-order policy	KHAHP-39.06A · KHAHP-39.06B · KHAHP-39.06C
39.07	Firmware update, DKMS/MOK and Secure-Boot readiness	KHAHP-39.07A · KHAHP-39.07B · KHAHP-39.07C

40 — Auditd / AIDE / Integrity Monitoring

Status: PROJECT ACCEPTED - COMPLETE / NO OPEN IMPLEMENTATION GAP.

Evidence position: FINAL ACCEPTANCE COMPLETE. In normal USER boot, auditd is enabled/active with the KHAHP rules effectively loaded, zero lost audit records in the acceptance capture, and persistent rule compilation verified. The production AIDE scheduled service completed successfully with exit status 0 after the dotlockfile permission repair; dailyaidecheck.timer remains enabled. SYSMAINT no-rules observations were correctly identified as boot-role behavior because auditd is inactive there.

Acceptance: All 21 A/B/C checkpoints for this section are accepted within the KHAHP 01-40 closeout. Re-test after material change; this does not imply third-party certification.

Family	Exact technical scope	A/B/C evidence IDs
40.01	auditd package/service state	KHAHP-40.01A · KHAHP-40.01B · KHAHP-40.01C

Family	Exact technical scope	A/B/C evidence IDs
40.02	Audit rule inventory and immutable/finalize policy	KHAHP-40.02A · KHAHP-40.02B · KHAHP-40.02C
40.03	Privileged-command and identity- change audit coverage	KHAHP-40.03A · KHAHP-40.03B · KHAHP-40.03C
40.04	AIDE package/database/configuration state	KHAHP-40.04A · KHAHP-40.04B · KHAHP-40.04C
40.05	File-integrity baseline and scheduled checks	KHAHP-40.05A · KHAHP-40.05B · KHAHP-40.05C
40.06	Audit-log ownership, rotation and tamper resistance	KHAHP-40.06A · KHAHP-40.06B · KHAHP-40.06C
40.07	CIS/DISA accountability-gap analysis	KHAHP-40.07A · KHAHP-40.07B · KHAHP-40.07C

41 — Cron / Timers / Persistence / Supply Chain

Status: NEXT PHASE - RECONNAISSANCE / IMPLEMENTATION NOT YET COMPLETE.

Evidence position: Persistence, scheduled activity and software supply-chain attribution form the first remaining final-assurance phase.

Required completion: Complete reconnaissance for this section, implement/accept applicable hardening decisions, then execute and accept its 21 A/B/C checkpoints before declaring the full 44-section program complete.

Family	Exact technical scope	A/B/C evidence IDs
41.01	cron/anacron job inventory	KHAHP-41.01A · KHAHP-41.01B · KHAHP-41.01C
41.02	systemd timer inventory	KHAHP-41.02A · KHAHP-41.02B · KHAHP-41.02C
41.03	User autostart and desktop persistence inventory	KHAHP-41.03A · KHAHP-41.03B · KHAHP-41.03C
41.04	Package repository and signing-key provenance	KHAHP-41.04A · KHAHP-41.04B · KHAHP-41.04C

Family	Exact technical scope	A/B/C evidence IDs
41.05	APT pinning/holds and update-policy review	KHAHP-41.05A · KHAHP-41.05B · KHAHP-41.05C
41.06	Unexpected startup scripts/services/modules	KHAHP-41.06A · KHAHP-41.06B · KHAHP-41.06C
41.07	Supply-chain drift and reboot- persistence evidence	KHAHP-41.07A · KHAHP-41.07B · KHAHP-41.07C

42 — Credentials / SSH Keys / Secrets / Backup

Status: NEXT PHASE - RECONNAISSANCE / IMPLEMENTATION NOT YET COMPLETE.

Evidence position: Credentials, secret handling, encrypted backup and isolated recovery form the second remaining final-assurance phase.

Required completion: Complete reconnaissance for this section, implement/accept applicable hardening decisions, then execute and accept its 21 A/B/C checkpoints before declaring the full 44-section program complete.

Family	Exact technical scope	A/B/C evidence IDs
42.01	SSH private-key inventory and permissions	KHAHP-42.01A · KHAHP-42.01B · KHAHP-42.01C
42.02	API/token/secret storage locations	KHAHP-42.02A · KHAHP-42.02B · KHAHP-42.02C
42.03	Password-manager selection and vault policy	KHAHP-42.03A · KHAHP-42.03B · KHAHP-42.03C
42.04	FIDO2/YubiKey readiness and hardware-backed keys	KHAHP-42.04A · KHAHP-42.04B · KHAHP-42.04C
42.05	Recovery-code generation and offline storage	KHAHP-42.05A · KHAHP-42.05B · KHAHP-42.05C
42.06	Encrypted backup creation and retention	KHAHP-42.06A · KHAHP-42.06B · KHAHP-42.06C

Family	Exact technical scope	A/B/C evidence IDs
42.07	Restore test, key recovery and disaster-recovery evidence	KHAHP-42.07A · KHAHP-42.07B · KHAHP-42.07C

43 — CIS / DISA GPOS / KSPP Mapping

Status: NEXT PHASE - RECONNAISSANCE / IMPLEMENTATION NOT YET COMPLETE.

Evidence position: Detailed CIS Debian 13 Level 2, DISA GPOS/SRG and KSPP applicability mapping remains to be completed without claiming external certification.

Required completion: Complete reconnaissance for this section, implement/accept applicable hardening decisions, then execute and accept its 21 A/B/C checkpoints before declaring the full 44-section program complete.

Family	Exact technical scope	A/B/C evidence IDs
43.01	CIS Debian 13 Level 2 applicability mapping	KHAHP-43.01A · KHAHP-43.01B · KHAHP-43.01C
43.02	DISA GPOS/SRG objective mapping	KHAHP-43.02A · KHAHP-43.02B · KHAHP-43.02C
43.03	KSPP/kernel-self-protection mapping	KHAHP-43.03A · KHAHP-43.03B · KHAHP-43.03C
43.04	Kicksecure-native equivalent control mapping	KHAHP-43.04A · KHAHP-43.04B · KHAHP-43.04C
43.05	Project-extension control mapping	KHAHP-43.05A · KHAHP-43.05B · KHAHP-43.05C
43.06	Exception/N-A/manual-control rationale	KHAHP-43.06A · KHAHP-43.06B · KHAHP-43.06C
43.07	Evidence citation and no-certification disclaimer	KHAHP-43.07A · KHAHP-43.07B · KHAHP-43.07C

44 — Final Acceptance / Evidence / Scoring

Status: NEXT PHASE - RECONNAISSANCE / IMPLEMENTATION NOT YET COMPLETE.

Evidence position: Whole-program regression, recovery, evidence-chain and final acceptance remain the final project phase.

Required completion: Complete reconnaissance for this section, implement/accept applicable hardening decisions, then execute and accept its 21 A/B/C checkpoints before declaring the full 44-section program complete.

Family	Exact technical scope	A/B/C evidence IDs
44.01	Full reboot persistence acceptance	KHAHP-44.01A · KHAHP-44.01B · KHAHP-44.01C
44.02	USER-mode regression acceptance	KHAHP-44.02A · KHAHP-44.02B · KHAHP-44.02C
44.03	SYSMAINT-mode regression acceptance	KHAHP-44.03A · KHAHP-44.03B · KHAHP-44.03C
44.04	Network/DNS/firewall/Tor/USB negative-test acceptance	KHAHP-44.04A · KHAHP-44.04B · KHAHP-44.04C
44.05	Recovery/rollback/backup acceptance	KHAHP-44.05A · KHAHP-44.05B · KHAHP-44.05C
44.06	Evidence completeness, hashes and chain-of-custody bundle	KHAHP-44.06A · KHAHP-44.06B · KHAHP-44.06C
44.07	Final scorecard, residual risk and operator sign-off	KHAHP-44.07A · KHAHP-44.07B · KHAHP-44.07C

6. Historical 01-38 action reconciliation - 63 records closed

The 63 inherited action records are retained for traceability. They are **not 63 vulnerabilities** and they are no longer an open implementation backlog. Under the 19 October 2026 project closeout they are resolved, accepted, superseded, or closed by an explicit policy/compatibility disposition within Sections 01-40.

ID	Sec.	Historical engineering issue	v5.0 disposition
U01-001	7	/tmp and /dev/shm noexec compatibility decision	**CLOSED / PROJECT ACCEPTED**
U01-002	7	/boot/efi-before-/boot fstab ordering review	**CLOSED / PROJECT ACCEPTED**
U01-003	2	permission-hardener package/provenance classifier	**CLOSED / PROJECT ACCEPTED**
U01-004	5	sync account shell classification	**CLOSED / PROJECT ACCEPTED**
U01-005	12	Unverified kernel tokens / CFI-KCFI claims	**CLOSED / PROJECT ACCEPTED**
U01-006	13	vesafb module applicability / dry-run review	**CLOSED / PROJECT ACCEPTED**
U01-007	14	USER versus SYSMAINT DKMS evidence reconciliation	**CLOSED / PROJECT ACCEPTED**
U01-008	8	Physical GRUB edit-denial test	**CLOSED / PROJECT ACCEPTED**
U01-009	6	LUKS key recovery and backup restore	**CLOSED / PROJECT ACCEPTED**
U01-010	10	Firmware/preboot DMA protection acceptance	**CLOSED / PROJECT ACCEPTED**
U01-011	7	Post-reboot SYSMAINT persistence evidence	**CLOSED / PROJECT ACCEPTED**

ID	Sec.	Historical engineering issue	v5.0 disposition
U01-012	7	/home, /var/tmp and noexec compatibility review	**CLOSED / PROJECT ACCEPTED**
U01-013	14	Signed-module negative test and update regression	**CLOSED / PROJECT ACCEPTED**
U01-014	7	/tmp nodev,nosuid hardening — implemented baseline	**CLOSED / PROJECT ACCEPTED**
U15-001	15	Strict Firefox AppArmor confinement is not established	**CLOSED / PROJECT ACCEPTED**
U15-002	15	AppArmor service policy and denial regression	**CLOSED / PROJECT ACCEPTED**
U15-003	15	USER versus SYSMAINT browser-label persistence	**CLOSED / PROJECT ACCEPTED**
U15-004	16	Privileged binary provenance/drift review	**CLOSED / PROJECT ACCEPTED**
U15-005	16	gst-ptp-helper capability policy/compatibility	**CLOSED / PROJECT ACCEPTED**
U15-006	17	setup-dist status-files 0777 exception monitoring	**CLOSED / PROJECT ACCEPTED**
U15-007	17	Sensitive directory/ACL/umask and tmpfs drift	**CLOSED / PROJECT ACCEPTED**

ID	Sec.	Historical engineering issue	v5.0 disposition
U15-008	18	faillock recovery and self-lockout policy	**CLOSED / PROJECT ACCEPTED**
U15-009	18	Passphrase/aging policy ownership	**CLOSED / PROJECT ACCEPTED**
U15-010	18	PAM USER/SYSMAINT interaction regression	**CLOSED / PROJECT ACCEPTED**
U15-011	19	Network SSH masking and listener persistence	**CLOSED / PROJECT ACCEPTED**
U15-012	19	Local SSH AF_UNIX socket and client key policy	**CLOSED / PROJECT ACCEPTED**
U15-013	20	Role-specific systemd service drift and listening sockets	**CLOSED / PROJECT ACCEPTED**
U15-014	20	Classify unexpected/unused services safely	**CLOSED / PROJECT ACCEPTED**
U15-015	21	Unbound systemd exposure score needs design decision	**CLOSED / PROJECT ACCEPTED**
U15-016	21	Service sandbox capability regression	**CLOSED / PROJECT ACCEPTED**
U15-017	22	Core dump/pstore suppression drift	**CLOSED / PROJECT ACCEPTED**
U15-018	22	Crash-forensics versus privacy trade-off	**CLOSED / PROJECT ACCEPTED**

ID	Sec.	Historical engineering issue	v5.0 disposition
U15-019	23	Journal retention, permissions and rotation policy	**CLOSED / PROJECT ACCEPTED**
U15-020	23	sdwdate time and journal chronology after reboot	**CLOSED / PROJECT ACCEPTED**
U15-021	24	Owner-approved panic/recovery policy and escape paths	**CLOSED / PROJECT ACCEPTED**
U15-022	24	Emergency shutdown and controlled recovery evidence	**CLOSED / PROJECT ACCEPTED**
U25-001	26	External USB keyboard possibly allowed	**CLOSED / PROJECT ACCEPTED**
U25-002	26	Composite/USB network rejection and persistence	**CLOSED / PROJECT ACCEPTED**
U25-003	25	Swap crypt mapping and shutdown key lifecycle	**CLOSED / PROJECT ACCEPTED**
U25-004	25	RAM wipe shutdown execution and hardware limits	**CLOSED / PROJECT ACCEPTED**
U25-005	27	Actual Wi-Fi MAC identity	**CLOSED / PROJECT ACCEPTED**
U25-006	27	DHCP identity rotation	**CLOSED / PROJECT ACCEPTED**
U25-007	28	Alternate IPv6 network profiles	**CLOSED / PROJECT ACCEPTED**

ID	Sec.	Historical engineering issue	v5.0 disposition
U25-008	29	Unbound response to DHCP renew/network loss	**CLOSED / PROJECT ACCEPTED**
U25-009	30	Unbound-owned authenticated DoT	**CLOSED / PROJECT ACCEPTED**
U25-010	30	Direct UDP/TCP 53 egress	**CLOSED / PROJECT ACCEPTED**
U25-011	30	Active browser DNS/DoH preferences	**CLOSED / PROJECT ACCEPTED**
U25-012	31	Effective firewall output chain	**CLOSED / PROJECT ACCEPTED**
U25-013	31	Exact loaded-ruleset persistence	**CLOSED / PROJECT ACCEPTED**
U25-014	32	Peer-side discovery proof	**CLOSED / PROJECT ACCEPTED**
U25-015	33	Tor Browser routing/DNS/fail-close	**CLOSED / PROJECT ACCEPTED**
U25-016	34	Secure-time unit chronology	**CLOSED / PROJECT ACCEPTED**
U25-017	25	Original implementation provenance	**CLOSED / PROJECT ACCEPTED**
U25-018	26	USB boot media usability versus detection	**CLOSED / PROJECT ACCEPTED**
U25-019	28	IPv6 goal mismatch / explicit policy	**CLOSED / PROJECT ACCEPTED**
U25-020	29	USER/SYSMAINT service role semantics	**CLOSED / PROJECT ACCEPTED**

ID	Sec.	Historical engineering issue	v5.0 disposition
U25-021	30	DNSSEC positive/negative evidence signoff	**CLOSED / PROJECT ACCEPTED**
U25-022	32	Actual scope of local discovery services	**CLOSED / PROJECT ACCEPTED**
U25-023	34	Production clock rollback exception	**CLOSED / PROJECT ACCEPTED**
U35-001	35	Early-boot entropy and random-seed lifecycle	**CLOSED / PROJECT ACCEPTED**
U35-002	36	Firefox effective DoH, WebRTC and telemetry behavior	**CLOSED / PROJECT ACCEPTED**
U35-003	37	Application history, thumbnails, metadata and confinement regression	**CLOSED / PROJECT ACCEPTED**
U35-004	38	Camera/microphone hotplug and real lock/lid-resume enforcement	**CLOSED / PROJECT ACCEPTED**

7. Commercial image engineering and supply-chain acceptance

The proposed KHAHP premium hardened OS remains a separate **release-engineering** workstream from the completed 01-40 laptop baseline. Any distributable ISO/USB, QCOW2/KVM or OVA image must be built cleanly and must not clone development-machine secrets, SSH material, random seeds or unique machine identities.

Clean-image common requirements

Release area	Required release proof
Reproducible source / package provenance	Rebuildable source chain, package-origin verification and isolated build process
Identity / credential sanitation	Unique machine identity, host keys and zero copied development credentials
Network / DNS / firewall defaults	Safe first-boot policy with documented recovery and customer configuration path
Kernel / MAC / audit / AIDE	Release-specific regression, not merely reuse of laptop evidence
Update / backup / restore	Supported update path, rollback and recovery validation
Release signing / documentation	Signed artifacts, hashes, licensing and operator guidance

Distribution-specific acceptance

- **Physical ISO / USB:** UEFI boot, fresh encrypted installation, recovery, Secure Boot, storage/device safety and two-device comparison.
- **KVM / QCOW2:** clean import/boot, unique clone identity, virtual firmware/TPM, snapshot restore, networking and sanitized release image.
- **VirtualBox / OVA:** clean import, identity regeneration, clipboard/shared-folder/USB/network policy, restore and host-version support.

8. Planned completion of Sections 41-44

Section	Remaining work	Checkpoints
41 - Persistence / supply chain	Cron/timers/autostart classification, package/signing provenance and drift	**21**
42 - Credentials / backup / recovery	Secret inventory, custody policy, encrypted backups and isolated restore	**21**
43 - CIS / DISA / KSPP mapping	Evidence-linked applicability and extension mapping; no certification inflation	**21**

Section	Remaining work	Checkpoints
44 - Final acceptance / evidence / scoring	Whole-program regression, recovery, chain of custody and final sign-off	**21**
Total remaining		**84**

9. Compliance, controlled change and public disclosure

Execution roles: normal USER activity remains separate from privileged SYSMAINT maintenance. Privileged work must preserve role boundaries and recovery paths.

Change discipline: evidence -> root cause -> minimal approved change -> backup/rollback -> postflight -> bounded negative test -> relevant persistence/recovery -> acceptance.

Publication safety: do not publish passwords, private keys, recovery material, private host identifiers, raw confidential logs, sensitive IP/MAC data or customer-specific security configuration. The public register publishes sanitized outcomes; underlying evidence remains private.

Reference frameworks: CIS Debian 13 Level 2 Workstation, DISA General Purpose OS SRG / relevant STIG-inspired objectives, KSPP, Debian and Kicksecure security practices. These are engineering references. **No third-party CIS, DISA or KSPP certification is claimed.**

10. Evidence source ledger for v5.0

- Public v4.2 README and full technical register, used as the structural baseline for this revision.
- KHAHP authoritative 924-gate ledger v3.0, 19 October 2026: Sections 01-40 / 840 gates recorded as `PROJECT_ACCEPTED`; Sections 41-44 / 84 gates remain the next reconnaissance, implementation and final-assurance phase.
- KHAHP master-control v4.0 and engineering-operations manual v2.0, 19 October 2026.
- Section 40 final evidence: USER/SYSMAINT delta reports, successful controlled production AIDE run, final root audit configuration capture, and normal-USER effective audit-rule capture.
- Section 39 remains owner-complete / no-retest unless intentionally reopened.

Section 40 final evidence summary

- `auditd.service`: enabled, active and running in normal USER boot during final acceptance.

- Kernel audit subsystem: enabled with live auditd PID; acceptance capture showed `lost 0`.
- KHAHP audit rules: persistent rule files compiled without change and effective in normal USER boot.
- SYSMAINT `auditctl -l: No rules`: correctly classified as role-specific because auditd is inactive in SYSMAINT.
- AIDE: existing production `dailyaidecheck.service` ran successfully after the lock-helper permission repair, with `ExecMainStatus=0` and successful completion.
- `/usr/bin/dotlockfile`: persistent `755 root:mail` override retained after the successful production run.
- External mail notification and audit immutability are treated as explicit policy/operational choices, not open implementation defects in the 01-40 closeout.

Release boundary: v5.0 records the 01-40 project closeout. It does not claim that Sections 41-44 are complete, that every commercial-image release test has been executed, or that an external benchmark authority has certified the system.